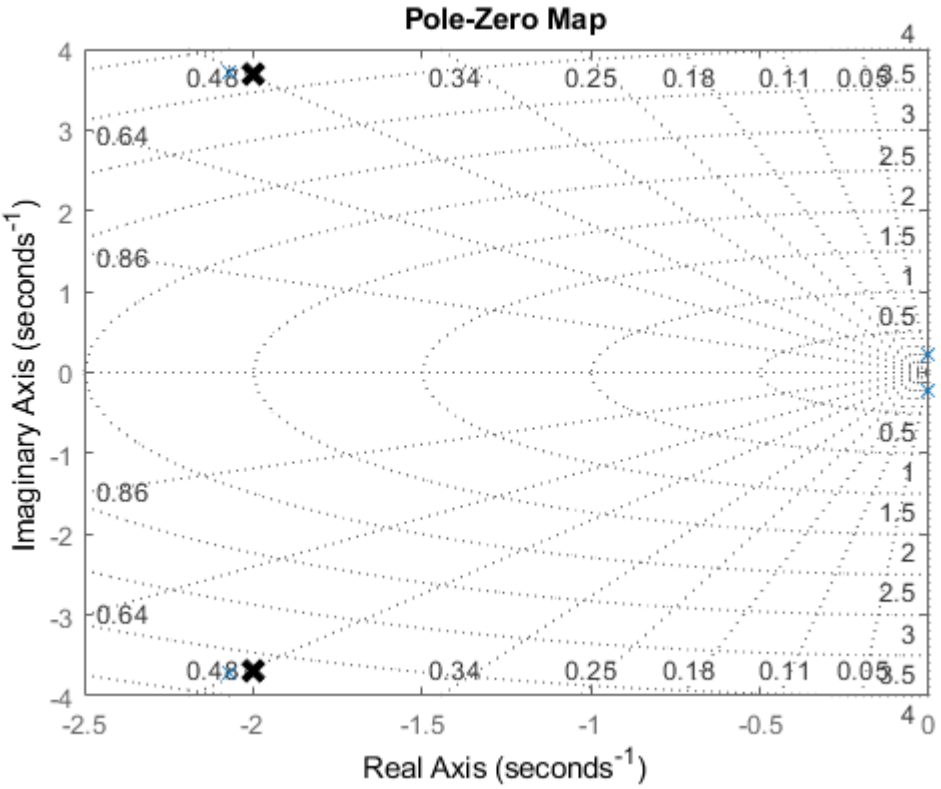
Exercise 10.1: Longitudinal Eigenmotion

Consider the following pole pair of an eigenmode of the longitudinal motion:



a) Which eigenmotion is displayed here?

b) Give an approximation for the derivatives, which define the eigenmode. You can neglect the influence of the pitch rate on the forces. Further simplifications have to be justified!

c) Does the pole zero plot in 10.a) belong to the following linear state space model of a Boeing 777?

$$\begin{pmatrix} \dot{V}^* \\ \dot{\gamma} \\ \dot{\alpha} \\ \dot{q} \end{pmatrix} = \begin{bmatrix} -0.01 & -0.49 & -0.046 & -0.001 \\ 0.11 & 0.0003 & 1.14 & 0.043 \\ -0.11 & -0.0003 & -1.14 & 0.957 \\ 0.1 & 0 & -15.34 & -3.00 \end{bmatrix} \begin{pmatrix} V^* \\ \gamma \\ \alpha \\ q \end{pmatrix} + \begin{bmatrix} 0.0002 & 0.014 \\ -0.059 & 0 \\ 0.059 & 0 \\ 6.56 & 0.13 \end{bmatrix} \cdot \begin{pmatrix} \eta \\ \delta_T \end{pmatrix}$$

Exercise 10.2: Lateral Eigenmotion

The following linear model of the lateral motion is given:

$$\begin{pmatrix} \dot{r} \\ \dot{\beta} \\ \dot{p} \\ \dot{\Phi} \end{pmatrix} = \begin{bmatrix} -0.6 & 6.0 & -0.18 & 0 \\ -1 & -0.19 & 0 & 0.13 \\ 1.35 & -15.10 & -6.5 & 0 \\ 0.05 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{pmatrix} r \\ \beta \\ p \\ \Phi \end{pmatrix} + \begin{bmatrix} 0.0075 & -2.6 \\ 0 & 0.037 \\ -13.09 & 1.87 \\ 0 & 0 \end{bmatrix} \cdot \begin{pmatrix} \xi \\ \zeta \end{pmatrix}$$

a) Calculate the initial roll acceleration and the steady state roll rate for an aileron deflection of 1°.

b) Plot the step response of roll rate and bank angle for the given aileron deflection.